

# Ulsan-Fault earthquake catalog pipeline

Reference manual: detection → association → location → relocation (HypoDD dt.ct + dt.cc)

Long-term (2010–present) catalog for the Ulsan Fault, SE Korea

rev. 2026-07-27 · src/ufpipe

`ufpipe` (`src/ufpipe`) is the **end-to-end pipeline**: six stages, detection → association → augment → PHS → locate → **relocate** (HypoDD dt.ct + dt.cc). The final **relocate** stage produces the double-difference relocated catalog. The heavy `dt.cc/xcorr/HypoDD engine` is external (15.PocketQuake korea-cluster-relocation, HypoDD v2.1beta); `ufpipe` orchestrates it. The relocate stage is **self-fed**: it builds its inputs from `ufpipe`’s OWN detection+association (`ufpipe.reloc_inputs`) and runs the orchestration in `src/ufpipe/reloc_driver/`; working dirs + results go to `outputs/reloc/`. `detection_test/` is a **fully frozen pilot archive** (the 2016 4-picker comparison; `lib/` DEPRECATED, see `detection_test/lib/DEPRECATED.md`). The relocation internals have a dedicated guide: `src/ufpipe/reloc_driver/PIPELINE.md` + the pilot’s `study_guide.pdf`.

**Data scope (verified 2026-07)**: the waveform archive covers **2010–2024**, ending at 2024 day 305 (31 Oct 2024). No 2025+ data is present in any network directory, so 2024’s 305 daily pick files are complete for the data that exists.

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## 1 What the pipeline does

Build an **internally consistent, long-term earthquake catalog** for the Ulsan Fault region (SE Korea), 2010–present, from continuous seismic waveforms, using AI phase pickers so small events are detected uniformly even as the recording network grows, and finally **relocate** the catalog with double-difference (HypoDD dt.ct + dt.cc) for sub-100 m relative precision. Every stage is scripted and parameterized; results reproducible.

| Detection            | Association            | Location           | QC                    | Relocation           |
|----------------------|------------------------|--------------------|-----------------------|----------------------|
| PhaseNet / PhaseNet+ | PyOcto (+augment)      | HYPONVERSE         | uf_cluster            | HypoDD dt.ct + dt.cc |
| <i>picks</i> →       | <i>events+assign</i> → | <i>.sum/.arc</i> → | <i>well-located</i> → | <i>hypoDD.reloc</i>  |

### 1.1 ufpipes vs detection\_test

ufpipe is the production end-to-end pipeline; everything live is under `src/`, including the relocation orchestrator `src/ufpipe/reloc_driver/run_picker_reloc.py` (driven by the relocate stage with `--skip-build`; the HypoDD/xcorr engine itself is external in 15.PocketQuake). `detection_test` is the **frozen pilot archive** of the 2016 4-picker comparison — nothing in it is live (`lib/` DEPRECATED; the `reloc_2016_uf*/` dirs hold the pilot’s manifests and results).

|         | ufpipe (src/ufpipe)                                                                             | detection_test                            |
|---------|-------------------------------------------------------------------------------------------------|-------------------------------------------|
| Role    | the self-contained end-to-end catalog + relocation pipeline (incl. <code>reloc_driver/</code> ) | frozen archive of the 2016 4-picker pilot |
| Stages  | detection → association → augment → PHS → locate → <b>relocate</b>                              | — (nothing live)                          |
| Driver  | <code>python -m ufpipes.run_pipeline</code>                                                     | —                                         |
| Ends at | HypoDD dt.cc (§7); results in <code>outputs/reloc/</code>                                       | —                                         |

ufpipe’s own detection + association cover **KS/KG/GJ/NS** with daily-chunked association (§5), and the **relocate** stage builds its inputs from those outputs (`ufpipe.reloc_inputs`); it no longer consumes `detection_test/lib`. The pipeline is fully self-contained. See §7.

### 1.2 Two orthogonal “model” dimensions

| Dimension             | Flag                    | Values                                                                                           |
|-----------------------|-------------------------|--------------------------------------------------------------------------------------------------|
| <b>Picker model</b>   | <code>--model</code>    | <code>stead</code> , <code>original</code> , <code>phasenet_plus</code> , <code>eqt</code> , ... |
| <b>Velocity model</b> | <code>--velmodel</code> | <code>kim1983</code> , <code>kim2011</code>                                                      |

Any picker model can be located with any velocity model. `stead/original` are SeisBench PhaseNet weights; `phasenet_plus` is EQNet PhaseNet+ (needs a local EQNet clone).

## 2 Install & environment

The pipeline runs in a **two-env split** on this server (there is no single “ulsan” env): detection (PhaseNet+/SeisBench, torch) in `eqnet` (Py 3.9); association (PyOcto) + orchestration in `base` (Py 3.12); the relocate stage’s GPU xcorr shells out to `pq-gpu` internally.

**One-time setup**

1. `conda run -n eqnet pip install -e . --no-deps` (detection env)
2. `conda run -n base pip install -e . --no-deps` (association/orchestration env)
3. **HYPOINVERSE**: the external `hyp1.40` binary must be on `PATH` (not pip-installable).
4. **PhaseNet+** only: a local EQNet clone at `config.EQNET_DIR`.
5. **Fresh machine**: `conda env create -f environment.yml` (env `uf-catalog`), then install the torch build matching your CUDA (`pip install torch --index-url https://download.pytorch.org/whl/cu128`).

After `pip install -e .`, `import ufpipes.config` and `python -m ufpipes.run_pipeline` work from any directory — no `sys.path` hacks.

### 3 Stages 1–5: detection through location

`STAGES = [detection, association, augment, phs, locate, relocate]` — run in order. Stages 1–5 (below) are functions in `src/ufpipe/core.py`; the relocate stage (6) is `src/ufpipe/relocate.py` and has its own §7. `run_pipeline.py` chains them per year.

| # | Stage              | In → Out                                            | What it does                                                                                                                                                                                                                               |
|---|--------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>detection</b>   | continuous mSEED → picks/<br>(per station-day)      | AI picker on each station-day across <b>KS/KG/GJ/NS</b> (per-year multi-network table, <code>stations.py</code> ); anti-alias >100 Hz, unify mixed rates, resample to 100 Hz, pick P/S above threshold. GPU. Idempotent (skips done days). |
| 2 | <b>association</b> | picks → <code>pyocto_*.csv</code> + assignment      | PyOcto octree associator groups picks into events (gate below), <b>daily-chunked</b> ( $\pm 150$ s/day, in-day origins) so it scales to the dense NS array. Coords from the multi-network table.                                           |
| 3 | <b>augment</b>     | assignment → augmented assignment                   | scan daily picks for orphans PyOcto missed, add to each event so HYPOINVERSE sees the full pick set. Backs up + overwrites in place; idempotent.                                                                                           |
| 4 | <b>phs</b>         | assignment → <code>UF{year}.phs</code>              | write the HYPOINVERSE phase file from the (augmented) assignment.                                                                                                                                                                          |
| 5 | <b>locate</b>      | <code>.phs</code> → <code>UF{year}.{sum,arc}</code> | run HYPOINVERSE ( <code>hyp1.40</code> ) with the chosen velocity model → located catalog + errors.                                                                                                                                        |

**Pipeline order matters**

`augment` runs *between* `association` and `phs`, so HYPOINVERSE sees the full augmented pick set and computes correct `GAP/DMIN/ERZ/RMS`. Do not reorder.

**3.1 Source of truth for pick↔event assignment**

The **PyOcto assignment (after augmentation)** is authoritative for which `(station, phase)` tuples belong to each event. Do NOT use a legacy time-window pick dump — when two events fall <60 s apart at the same station the earliest pick “wins” and leaks into the wrong event. Downstream (SAC-header writers, magnitude/QC notebooks) must read `pyocto_assignment_{velmodel}_{year}.csv`.

**4 Parameters (src/ufpipe/config.py — the single disclosed source)**

Change defaults here. Everything is one file.

**4.1 Detection / DSP**

| Parameter                                              | Value                               | Meaning                                                     |
|--------------------------------------------------------|-------------------------------------|-------------------------------------------------------------|
| <code>SAMPLING_RATE</code>                             | 100.0 Hz                            | the pickers’ trained rate; all data resampled to it         |
| <code>ANTIALIAS</code>                                 | lowpass 45 Hz, 8 corners, zerophase | applied BEFORE down-sampling (>100 Hz KG/NS)                |
| <code>BANDPASS</code>                                  | 1–40 Hz, 4 corners, causal          | legacy detection path; PhaseNet+ runs raw                   |
| <code>P_THRESHOLD</code> ,<br><code>S_THRESHOLD</code> | 0.2                                 | SeisBench pick probability threshold                        |
| <code>PNPLUS_MIN_PROB</code>                           | 0.3                                 | EQNet PhaseNet+ pick threshold                              |
| <code>PNPLUS_NT</code>                                 | 1024×36                             | PhaseNet+ patch length (whole-day, <code>cut_patch</code> ) |
| <code>MAX_CORES</code>                                 | 24                                  | CPU cap (respects <code>taskset</code> affinity)            |

**4.2 Association (PyOcto) — daily-chunked, ASSOC\_GATE (lenient) vs ASSOC\_GATE\_STRICT**

Association runs **one calendar day at a time**: a  $\pm$ `ASSOC_OVERLAP_S` (150 s) window per day is associated and only events whose *origin* falls in-day are kept (dedup). This is physically equivalent to a single pass (local events are seconds long) but keeps each PyOcto solve to seconds — a whole-year single pass is intractable on the dense ~200-station NS array ( $\gg$ 1 h, 12 GB). Station coordinates come from the multi-network year table (`stations.py`), so KS/KG/GJ/NS all associate.

**NS station epochs — moved stations are SPLIT and association is the rename boundary**

22 of the 220 GHBSN (NS) stations were physically relocated during operation (12m–5.2km; N108 and N166 twice). Each position epoch is its OWN station: lowercase suffixes from the *first* epoch (N010a, N010b; a plain code always means never-moved). The canonical epoch table (`ufpipe.ns_epochs`, generated from the user-validated SOTA list `stations/ns/20240715/GHBSN_info_ver202312_modified.csv`) is the single source; the old list carried a 13km paste error at N019 and the epochs were previously AVERAGED into fictitious midpoints. Detection and the raw picks always use BASE codes (position-free; the archive stores base-code dirs); at the association input, each NS pick is stamped with the epoch code in force at its pick time, and every downstream product — assignment, `.phs`, `.sta`, `station.dat`, SAC filenames and headers — knows only epoch codes. `xcorr` therefore never pairs waveforms across a move (correct: the differential geometry is broken by it), and responses alias epoch → base code (a move is not a re-equip). The NS\_100hz mirror is retired (it was a partial 2021 test build, 8% coverage, and its per-station swap silently dropped stations in other years); NS is read native 200 Hz with the same anti-alias path as GJ.

| Parameter (config)   | lenient                      | _STRICT | Meaning                            |
|----------------------|------------------------------|---------|------------------------------------|
| REGION_CENTER ± pad  | (35.856,129.224) ± (1.0,1.2) |         | search area (deg)                  |
| ASSOC_ZLIM           |                              | (0, 30) | depth range (km)                   |
| ASSOC_TIME_BEFORE    |                              | 300     | origin-search window (s)           |
| ASSOC_OVERLAP_S      |                              | 150     | daily-chunk overlap (s)            |
| n_picks              | 4                            | 6       | min total picks                    |
| n_p / n_s            | 2 / 2                        | 3 / 3   | min P / S picks                    |
| n_ps                 | 1                            | 2       | min stations with both P+S (depth) |
| ASSOC_PICK_MATCH_TOL |                              | 1.5     | residual cap (s)                   |

Every value above is **overridable per run** (defaults = config) with matching flags on `association.py` and `run_pipeline.py`: `--gate n_picks=..., n_p=..., n_s=..., n_ps=..., --pick-match-tol, --overlap-s, --zlim Z0,Z1, --center LAT,LON, --lat-pad, --lon-pad, --time-before; --strict` selects `ASSOC_GATE_STRICT` (ignored if `--gate` is given). The cockpit notebook sets the same knobs via `run_association_year(overrides=...)`. Velocity = `ASSOC_VELMODEL` (kim2011 1-D). **Station-code alias:** a picked code missing from the year table but present as `<code>2` (e.g. BUS→BUS2 — same station, renamed code) is remapped so its picks are used, not dropped. Output schema (`events idx,time,x,y,z,picks,latitude,longitude,depth`; assignment `event_idx,pick_idx,residual,station,phase,time`) is unchanged, so `phs/locate/relocate` consume it as before.

**HYPOINVERSE pick weights (stage 4 input).** `make_phs` sets the HYPO71 weight code from the picker's own probability (`config.PHS_WEIGHT_SCHEME="probability"`). Design principle (revised 2026-07): *association is the gatekeeper; the weight is only a prior* — every associated pick gets a NONZERO weight, and genuinely bad picks are killed by HYPOINVERSE's own distance + residual re-weighting, which sees the actual misfit. `PHS_WEIGHT_BINS` = even quartiles of the emitted range [0.2, 1.0]:  $\geq 0.80 \rightarrow 0$ ,  $\geq 0.60 \rightarrow 1$ ,  $\geq 0.40 \rightarrow 2$ , else  $\rightarrow 3$  (0 = full weight; **code 4 is never assigned**). S picks one code worse (`PHS_WEIGHT_S_PENALTY`), **capped at 3**. Probabilities are matched to picks by nearest time within `PHS_PROB_MATCH_TOL_S` (50 ms), alias-aware (BUS/BUS2); a pick with no match gets the conservative `PHS_WEIGHT_UNMATCHED_CODE=3`, not full weight. Set `PHS_WEIGHT_SCHEME="phase"` for the legacy fixed P=0/S=1.

The previous scheme ( $\geq 0.90/0.70/0.50/0.30$ , else 4, S+1 uncapped) discarded **17.5%** of associated picks (26% of S) and matched only 74% of probabilities, defaulting the rest to FULL weight; on 2016 it also produced the 92 all-weight-4-P events that segfaulted `hyp1.40`. The revision took 2016 from 11,807 located / 7,830 QC-pass to **13,123 located (99.9% of associated) / 9,440 QC-pass**, with median RMS unchanged (0.05 s).

### 4.3 Velocity model & location

DEFAULT\_VELMODEL = kim2011; kim1983 also available (data/metadata/velocity/kim1983.csv, a layered  $V_p/V_s$  model; HYPOINVERSE \*.crh tables live in data/hypoinv/{kim1983,kim2011}/). PHASE\_CHANNELS = {P: HHZ, S: HHN}.

The HYPOINVERSE station file is **GENERATED, not hand-maintained**. Before locating, `stations.write_hypoinverse_sta(year)` derives `STA/UF<year>_hyp.sta` (fixed-format) and `STA/UF<year>.sta` (CSV, read by `uf_cluster.load_stations`) from the multi-network year table — the single source of truth. `--no-regen-sta` keeps an existing file; the previous hand file is preserved once as \*.legacy.

#### Why this matters — a silent and destructive failure mode

A stale or incomplete station list does *not* raise an error. HYPOINVERSE prints SKIP PHASE CARD WITH UNKNOWN STATION for every pick at a station it does not know, then drops any event left with < 4 usable phases (CANT SOLVE) — and those losses propagate into the `.arc` → `ph2dt` → `event.dat/dt.ct` that HypoDD relocates on. The shipped legacy list held **10** stations for 2010 while the association used **12**: 21% of picks were discarded. Regenerating took 2010 from **640 located / 161 QC-pass** to **791 / 303**. The gap was far larger in later years — the legacy list knew 13 of 46 stations in 2016 (no GJ array) and 44 of 243 in 2021 (no NS array).

### 4.4 Instrument responses — all four networks (src/ufpipe/responses.py)

`responses.py` is to responses what `stations.py` is to coordinates: the single source of truth. `load_inventory(networks=None)` returns one merged `obspy.Inventory` spanning KS/KG/GJ/NS, built master-first so an authoritative entry is never shadowed by a per-station fill. Sources, all under `data/metadata/responses/`:

| Network      | Directory          | Format     | Coverage                        |
|--------------|--------------------|------------|---------------------------------|
| KS + KG      | master/            | StationXML | KS_KG_metadata_1.0.2.xml        |
| KS (7 gaps)  | fetched/extracted/ | SEED RESP  | fetched from NECIS              |
| GJ           | gj/                | SEED RESP  | 30 stations × 3 (MARA)          |
| NS           | ns/                | SEED RESP  | 200 stations × 3 (MARA)         |
| NS N201–N220 | ns_derived/        | SEED RESP  | 20 stations × 3, <b>derived</b> |

Coverage is **100% of the station table for every year 2010–2024**, all three channels. Check it with `python -m ufpipes.responses --coverage <year>`. The ML side reaches the same inventory through `ml_pipeline.load_full_inventory()`.

### A missing response is silent rather than loud

Response removal is what turns counts into Wood-Anderson millimetres. A station absent from the inventory raises no error — `remove_response` simply skips it, so the event’s ML is the median over whatever stations happened to have metadata. Before GJ and NS responses existed, an ML computed on a 2016 or 2021 catalogue would have quietly used only the KS/KG minority (7 of 46 stations in 2016; 43 of 243 in 2021) while appearing to succeed. Run the coverage check before trusting any ML catalogue.

### Renamed stations — why the response lookup is epoch-aware

KMA renames a station when it is re-installed: BUS → BUS2 → BUS3, DAG → DAG2, USN → USN2 — *identical coordinates*, strictly sequential response epochs. `responses.epoch_families()` discovers these from the StationXML coordinates (never from the trailing digit, so GKP1 and N001 are safe) and requires the epochs to be **non-overlapping**, which correctly excludes co-located *contemporaries* such as CHS/CS0 and INC/INCA — genuinely different instruments that must never be aliased. 23 families, 49 codes.

This matters because of one archive quirk: KS\_KG/BUS2/ holds **both** epochs’ data, and its miniSEED headers read BUS until 2015 — long after the Nov-2010 rename. Association rewrites those picks to BUS2, which is right for *position* (identical coordinates, so location and HypoDD are unaffected) but wrong for the *response*, whose epochs are strict. Without the epoch-aware lookup, 89% of 2010’s BUS picks silently lost their response. An archive-wide header audit found **this is the only station affected** — DAG2, ULJ2, USN2, GKP1/2, BUS3 and all GJ/NS dirs have headers matching their directory.

### ns\_derived/ rests on an assumption — N2xx magnitudes are provisional

The MARA dataless (2020-06-03) predates the N201–N220 deployment (first data 2022.237), so those 20 stations have no authoritative response. All 600 authoritative NS RESP files are byte-identical apart from station code and epoch start date — one instrument configuration for the whole array ( $754.3 \times 4 \times 10^5 = 3.0172 \times 10^8$  counts/(m/s), exact) — so `write_derived_ns` clones it for the missing block, stamping each with that station’s own first-day-on-disk as the start date. The assumption is that the 2022 expansion kept the same configuration; the N2xx stations do record at the same 200 Hz, in 3 components, at comparable count amplitudes, but there is no manufacturer record here to confirm it. Every generated file carries a DERIVED by `ufpipe.responses` header. Delete the directory and re-derive the moment real metadata arrives.



## 5 How to run

### Notebooks — one tidy folder per year

`python notebooks/make_year_notebooks.py --years 2010-2024` builds, for each year, `notebooks/<year>/00.Run_pipeline_<year>.ipynb` (the 6-stage cockpit: run + check + PyGMT maps, every knob in the top parameters cell) and `01.Record_sections_<year>.ipynb` (record sections with associated P/S picks + a dedicated augmented-events figure), with YEAR/MODEL pre-filled.

`python notebooks/make_year_notebooks.py --status` prints the per-year progress table (picks / assoc / located / QC / dt.cc) read from disk — the way to manage a multi-year run. Kernel base; the emitted notebooks are gitignored (the builders are tracked).

### Standard runs (from the repo root)

|                                                                                                                |                                       |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <code>python -m ufpipes.run_pipeline --model original --years 2024</code>                                      | one year, all 6 stages                |
| <code>python -m ufpipes.run_pipeline --model original --years 2010-2024</code>                                 | the whole record                      |
| <code>python -m ufpipes.run_pipeline --model original --years 2015,2016 --velmodel kim1983</code>              | specific years, kim1983               |
| <code>python -m ufpipes.run_pipeline --model original --years 2024 --stage-from association</code>             | resume (picks already done)           |
| <code>python -m ufpipes.run_pipeline --model original --years 2024 --days 1-3</code>                           | detection on a few days (testing)     |
| <code>python -m ufpipes.run_pipeline --model original --years 2024 --strict</code>                             | strict association gate               |
| <code>python -m ufpipes.run_pipeline --model original --years 2016 --stage-from relocate --through dtcc</code> | relocation only (dt.cc; needs feeder) |

**CLI flags:** `--model` · `--years` ('2010-2024' or '2015,2016') · `--velmodel` · `--days` · `--device` {cuda,cpu} · `--workers` · `--strict` · `--force` (allow writing to *stead*) · `--stage-from` {detection,association,augment,phs,locate,relocate} (start mid-chain) · `--gate/--pick-match-tol/...` (association, §4.2) · `--qc/--xcorr` (relocation, §7) · `--no-regen-sta` (keep an existing station file).

### Shared 64-core server — always pin cores

Detection sizes its preprocessing pool and torch threads from the process CPU *affinity*, capped by `MAX_CORES=24`. Launch under `taskset` so the whole job stays within budget:

```
OMP_NUM_THREADS=1 taskset -c 0-7 python -m ufpipes.run_pipeline --model original --years 2010-2024
```

Without pinning, PhaseNet+ grabbed ~49 cores / 193 threads and starved everything. Inference is **GPU-preferred** (warns loudly, never silently falls back to CPU).



Detection is **idempotent** — it skips station-days whose picks already exist, so runs resume safely after an interruption.

## 5.1 Running each stage separately

Every `ufpipe` stage is also a standalone CLI (`--model --year`), so you can run detection alone, then association alone, etc. Order is `detection` → `association` → `augment` → `phs` → `locate`. Each reads the previous stage's output from disk.

| Stage       | standalone command                                                                                                                        |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Detection   | <code>python -m ufpipes.detection --model original --year 2024</code><br><code>[--networks KS,KG,GJ,NS] [--days 1-5] [--workers 8]</code> |
| Association | <code>python -m ufpipes.association --model original --year 2024</code><br><code>[--networks KS,KG,GJ,NS] [--workers 8] [--strict]</code> |
| Augment     | (runs inside <code>run_pipeline</code> ; or call <code>core.run_augment_year</code> from Python)                                          |
| PHS         | <code>python -m ufpipes.make_phs --model original --year 2024</code>                                                                      |
| Location    | <code>python -m ufpipes.run_hypoinverse --model original --year 2024</code><br><code>--velmodel kim2011</code>                            |

Or drive the same stage functions from Python / a notebook:

```
from ufpipes import core, config
core.run_detection_year("original", 2024, days=range(1,6))
events, assign = core.run_association_year("original", 2024)
core.write_phs("original", 2024); core.run_hypoinverse_year("original", 2024,
velmodel="kim2011")
```

## 5.2 Relocation — running the last stage (stage 6)

The relocate stage is **self-fed**: it needs only `ufpipes`'s own per-year detection + association (`pyocto_kim2011_<year>.csv` + assignment). It builds the reloc input store (event-idx SAC + station table) from those via `ufpipes.reloc_inputs`, then hands off to the driver `run_picker_reloc.py` `--skip-build` (scaffold → HYPOINVERSE → QC → rereference → GPU `xcorr` → HypoDD `dt.cc`). If a year's association is missing, the stage preflights and prints exactly what to run.

| Step          | command                                                                                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Detection     | <code>conda run -n eqnet python -m ufpipes.detection --model original</code><br><code>--year 2016</code>                                                                          |
| Association   | <code>conda run -n base python -m ufpipes.association --model</code><br><code>original --year 2016</code>                                                                         |
| Reloc (to QC) | <code>python -m ufpipes.run_pipeline --model original --years 2016</code><br><code>--stage-from relocate --through hypoinverse (fast)</code>                                      |
| Reloc (full)  | <code>python -m ufpipes.run_pipeline --model original --years 2016</code><br><code>--stage-from relocate --through dtcc (+ xcorr→HypoDD dt.cc;</code><br><code>~6 h dense)</code> |

`--through hypoinverse` stops at the QC'd absolute-location subset; `--through dtcc` continues to the double-difference relocation. `--clean-cache` deletes the ~tens-of-GB interp cache after `dt.cc` (essential for multi-year runs). Two override flags expose the stage-6 knobs per run (defaults = the validated values): `--qc "erh=5,erz=5,gap=270,num=5,rms=1.0"` (the `uf_cluster.QC` gate) and `--xcorr`

"interp\_hz=1000,fmin=5,fmax=20,cc\_threshold=0.7,pre=0.5,post=0.5,margin=0.5" (the dt.cc cross-correlation). The HypoDD adaptive-damping/ISTART=2 scheme and the full-run-HYPOINVERSE injection are fixed invariants (not flag-settable). The cockpit notebook (notebooks/00.Run\_yearly\_pipeline.ipynb) sets all of these from its parameters cell. `detection_test/lib/` is DEPRECATED (see `lib/DEPRECATED.md`); the relocation internals are documented in `src/ufpipe/reloc_driver/PIPELINE.md` + `study_guide.pdf`.

## 6 Outputs & layout

All pipeline products go under `outputs/models/<model>/` (regenerable; gitignored):

| Path                                                          | Contents                                           |
|---------------------------------------------------------------|----------------------------------------------------|
| <code>.../detection_location/&lt;year&gt;/picks/</code>       | per-station-day picks (stage 1)                    |
| <code>.../pyocto/pyocto_{vm}_{year}.csv</code>                | associated events (stage 2)                        |
| <code>.../pyocto/pyocto_assignment_{vm}_{year}.csv</code>     | pick→event assignment (authoritative)              |
| <code>.../station_table/stations_{year}.csv</code>            | per-year station roster (discovered from metadata) |
| <code>.../HypoInv/PHS/UF{year}.phs</code>                     | HYPONVERSE phase file (stage 4)                    |
| <code>.../HypoInv/{vm}/UF{year}.{sum,arc,prt}</code>          | located catalog + errors (stage 5)                 |
| <code>outputs/reloc/reloc_{year}_uf[_{model}]/results/</code> | dt.cc-relocated catalog + links (stage 6)          |

`{vm}` = `config.PYOCTO_VELMODEL` = **kim2011** for current daily-chunked runs; legacy pre-2026-07 whole-year artifacts on disk are named **kim1983**.

**Inputs** (metadata by kind, 2026-07): waveforms in `KS_KG/ GJ/ NS/ NS_100hz/` (station dirs at each root); station tables in `data/metadata/stations/{ks_kg,gj,ns,kigam}/` (+ the per-year derived cache `stations/derived/`); instrument responses for all four networks in `data/metadata/responses/{master,fetched,gj,ns,ns_derived}/` (§4.4); velocity model in `data/metadata/velocity/`; HYPONVERSE control inputs (STA/kim\*) in `data/hypoinv/`.

## 7 Stage 6: relocation — HypoDD dt.ct + dt.cc

The 6th ufpipes stage: turn the QC'd absolute-location catalog into a **double-difference relocated** catalog for sub-100 m relative precision.

```
python -m ufpipes.run_pipeline --model original --years 2016 --stage-from relocate
--through dtcc --clean-cache
```

`ufpipes.relocate` (`src/ufpipes/relocate.py`) is *self-fed*: it preflights ufpipes's OWN per-year association, builds the reloc inputs from it (`ufpipes.reloc_inputs` — SAC store, pyocto tables, station table, merged archive), then invokes the validated orchestrator `src/ufpipes/reloc_driver/run_picker_reloc.py --skip-build`, which drives the external HypoDD/xcorr engine. Internal stages (all KS/KG/GJ/NS):

| # | Stage                                     | What it does                                                                                                                                                |
|---|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | SAC store                                 | event-idx-keyed native-rate SAC waveforms from the associated picks ( <code>event_sac/</code> )                                                             |
| 2 | HYPOINVERSE                               | absolute location of all UF-box events (kim2011) → <code>.sum</code>                                                                                        |
| 3 | QC                                        | <code>uf_cluster.QC</code> : keep <code>erh&lt;5</code> , <code>erz&lt;5</code> , <code>gap&lt;270</code> , <code>num&gt;5</code> , <code>rms&lt;1.0</code> |
| 4 | inject full<br><code>.sum/.arc/STA</code> | QC subset reuses the full-run HYPOINVERSE solution + station file (no redundant re-run)                                                                     |
| 5 | ph2dt / dtct                              | build <code>event.dat</code> + catalog differential times <code>dt.ct</code> from the injected <code>.arc</code>                                            |
| 6 | re-reference                              | stamp each SAC's origin from the (full-run) HYPOINVERSE <code>.sum</code>                                                                                   |
| 7 | xcorr (GPU)                               | cross-correlation differential times <code>dt.cc</code> (band 5–20 Hz, interp 1000 Hz, <code>cc≥0.7</code> )                                                |
| 8 | HypoDD                                    | kim2011, ISTART=2, ISOLV=2 (LSQR), adaptive damping (CND 60–80), HypoDD v2.1beta (MAXDATA=15M) → <code>hypoDD.reloc</code>                                  |

#### Relocation outputs (symlinked into the working dir)

Each run publishes `outputs/reloc/reloc_<year>_uf[_<p>]/results/` with symlinks to the external run tree: `hypoDD.reloc.dtcc` (`dt.ct+dt.cc` — **the deliverable**), `HypoInv.full.sum`, `HypoInv.qc.sum`, ...“`dt.cc-resolved`” = events keeping  $\geq 1$  surviving cross-correlation link — the highest-precision subset. `hypoDD.reloc.dtct` (`dt.ct`-only) is an *optional diagnostic baseline*: it re-runs HypoDD on catalog differential times only (heavier, no `cc`) and is non-fatal + time-bounded, so it may be absent when slow — the `dt.cc` result is unaffected.

#### The invariant that matters (see PIPELINE.md)

Origins flow into `dt.cc`: `rereference` stamps SAC origins from a `.sum`, and `xcorr` stores  $dt = (t_1 + \text{shift} - ot_1) - (t_2 - ot_2)$  — subtracting those origins. Use the *full-run* HYPOINVERSE solution QC gated on (never a redundant re-run), or the `dt.cc` *values* are corrupted. This is enforced by `run_picker_reloc.py::inject_full_hypoinverse`.

## 8 Robustness & invariants

- **Network growth is automatic.** Stations are *discovered from StationXML + on-disk coverage* each month; the deployed network for that period is what enters the run. No hard-coded station lists.
- **Mixed sampling rates handled.**  $>100$  Hz stations (KG/NS at 200 Hz) are anti-alias-lowpassed then resampled to the pickers' 100 Hz. The anti-alias filter is applied BEFORE down-sampling (never after).
- **Velocity model, thresholds, and gates are epoch-invariant by choice**  $\Rightarrow$  the catalog stays internally comparable across years.
- **Idempotent & resumable** at every stage; `--stage-from` restarts mid-chain.

- **Do not edit the `stead` reference run** (it is the baseline). Scripts refuse `--model stead` writes unless `--force`.

## 9 Troubleshooting / provenance checks

| Symptom                                             | Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Detection finds no waveforms                        | <code>python -c "import ufpipes.stations as s; print(s.build_year_table(YEAR).net.value_counts())"</code><br>— the per-year multi-network table must list stations, and each row's <code>archive</code> dir must exist.                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>ImportError: ufpipes / uflib</code>           | <code>pip install -e . --no-deps</code> in <b>both</b> <code>eqnet</code> and <code>base</code> ; <code>import ufpipes.config; print(__file__)</code> must be <code>src/ufpipes</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>ModuleNotFoundError: pyocto</code>            | you ran association in <code>eqnet</code> — PyOcto lives in <code>base</code> .<br>Detection → <code>eqnet</code> ; association → <code>base</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>InternalMSEEDWarning: ...Steim2 failed</code> | corrupt Steim2 record(s) in the archive (e.g. KG.MKL 2010): the samples still decode; the reader collapses the per-record spam into one summary line per station-day and reads anyway. Informational — the named station-days are the QC signal.                                                                                                                                                                                                                                                                                                                                                                                    |
| GJ/NS station silently yields no picks              | mid-day sampling-rate changes (100/200/1000 Hz) and <code>int32/float64</code> records refuse to merge — handled since 2026-07 (rate + dtype unified <i>before</i> merge in both readers).                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>relocate</code> aborts at preflight           | ufpipes's own per-year association is missing for that (model, year) — run <code>ufpipes.detection</code> ( <code>eqnet</code> ) then <code>ufpipes.association</code> ( <code>base</code> ); the preflight prints the exact commands.                                                                                                                                                                                                                                                                                                                                                                                              |
| Wrong pipeline imported                             | ufpipes is renamed precisely to avoid the collision with the external korea-cluster-relocation pipeline. Verify <code>ufpipes.config.__file__</code> is under <code>src/</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Events look over/under-associated                   | <code>ASSOC_GATE</code> vs <code>--strict</code> ( <code>ASSOC_GATE_STRICT</code> ); <code>ASSOC_PICK_MATCH_TOL</code> is the primary knob.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| GPU not used                                        | inference is GPU-preferred and warns loudly; check the torch/CUDA install (never silently CPU).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Sparse early years, few events                      | expected — KS/KG-only network (2010–2015) has larger azimuthal gaps; GJ joins 2016, the dense NS array 2017+; the QC gate honestly reflects what the network resolves.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>.sum</code> stops mid-year; months missing    | <code>hyp1.40 segfaulted</code> partway (2016 once died on 25 Jun, dropping the whole Gyeongju sequence): it crashes in <code>hytrl_</code> on any event whose P picks are ALL weight code 4 (only possible under the probability weight scheme, every P below the last <code>PHS_WEIGHT_BINS</code> threshold). Fixed 2026-07: <code>write_phs</code> skips such unlocatable events (logged to <code>PHS/UF&lt;year&gt;.phs.skipped</code> ) and <code>run_hypoinverse_year</code> now raises on a nonzero exit code instead of shipping the partial <code>.sum</code> . If you see this, regenerate from stage <code>phs</code> . |

---

|                                                        |       |                   |                      |                      |
|--------------------------------------------------------|-------|-------------------|----------------------|----------------------|
| Companion                                              | docs: | docs/overview.md, | docs/pipeline.md,    | docs/how-to-run.md,  |
| docs/directory-structure.md                            |       | ·                 | src/ufpipe/README.md | ·                    |
| src/ufpipe/reloc_driver/PIPELINE.md + study_guide.pdf. |       |                   |                      | relocation pipeline: |